

Robust AI-Driven Intrusion Detection and Defense for Next-Generation Consumer Services

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Abstract—With the deep integration of 6G, the Internet of Things, and artificial intelligence, this paper proposes an intrusion detection and defense framework that combines robust AI kernel reconstruction, a cross-layer collaborative perception architecture, and a dynamic defense closed-loop mechanism to address advanced persistent threats and dynamically evolving attacks targeting next-generation consumer services. First, a lightweight detection model ATF-KDBC is designed based on adversarial training and online knowledge distillation. Gradient masking and noise injection are employed to enhance robustness against adversarial samples, while a drift-aware module enables adaptive optimization under concept drift scenarios. The model achieves accuracies of 99.25% and 99.84% on the NSL-KDD and IoT-23 hybrid datasets, respectively, and compresses the model size to 1.08 MB, representing a 97.6% reduction compared with the BERT teacher model. Second, a multidimensional attack chain analysis model is developed based on a STHGN. By integrating semantic, structural, and temporal features with a multi-head self-attention mechanism, the model enables cross-layer threat tracing and millisecond-level response, achieving an F1-score exceeding 97.0% on the DARPA dataset. Furthermore, this study explores the construction of a distributed CTIS network by integrating federated learning and blockchain technology. Zero-knowledge proofs are employed to ensure privacy preservation, while a Quality of Data and Quality of Model scoring mechanism enables efficient and precise deployment of defense strategies. Experimental results demonstrate that the proposed framework significantly outperforms traditional methods in terms of robustness, environmental adaptability, and computational efficiency, thereby providing both theoretical support and a technical pathway for enhancing the resilience and security of next-generation consumer services.

Index Terms—Artificial intelligence, consumer services, knowledge distillation, sixth-generation communication networks.

I. INTRODUCTION

DRIVEN by the rapid development of 6G and the Internet of Things (IoT), along with the deep integration of cloud computing, artificial intelligence, and big data analytics, the consumer services sector is entering a new stage of digital

transformation. Next-generation consumer services mark a paradigm shift from traditional online services, characterized by hyper-personalization, real-time interactivity, intelligent automation, and seamless omnichannel experiences. These services have deeply penetrated critical modern-life scenarios—such as real-time financial transactions, intelligent customer delivery, context-aware delivery platforms, and immersive social commerce [1]. This trend indicates that we are entering a new era of digital consumption in which virtually anything can be purchased online as a service via consumers' electronic devices. However, the combination of ubiquity, technical complexity, and operational criticality renders next-generation consumer services high-value targets for Advanced Persistent Threats (APTs) [2]. Security vulnerabilities in consumer electronics are evident through compromised smart home cameras enabling unauthorized disclosure of personal privacy, wearable IoT devices susceptible to unauthorized access of health data, and in-vehicle telematics systems subjected to remote control exploitation, with instances demonstrating critical security flaws in connected devices [3]. Ensuring their security, privacy, and resilience has moved beyond purely technical concerns to become a fundamental prerequisite for sustainable development and the establishment of user trust in the digital era.

The cloud-native architectures, API-driven economy, and deep AI integration underpinning next-generation consumer services, while fostering business innovation, have also introduced multi-layer and cross-layer coupled security challenges [4], [5], [6]. At the user-interaction layer, automated crawlers that exfiltrate sensitive data and adversarially crafted phishing attacks have resulted in annual losses exceeding \$10 billion. Lateral infiltration into business microservices and fraud targeting distributed transactions are further amplified and can proliferate rapidly in decentralized architectures. Targeted attacks on the AI-model layer are particularly severe because they directly exploit vulnerabilities in conventional defense mechanisms. [7], [8] while container escapes and supply-chain poisoning at the infrastructure layer further magnify systemic risk. These threats are dynamically evolving, coordinated across layers, and can produce nonlinear escalations of harm, thereby revealing fundamental limitations of detection schemes that depend on static rules and shallow AI—particularly with respect to robustness, correlation analysis, and real-time responsiveness. There is an urgent need for a new AI-driven detection-and-defense paradigm that integrates adversarial robustness, cross-layer perception, and adaptive decision-making [9].

Conventional intrusion-detection mechanisms exhibit three structural deficiencies when confronted with the dynamic threat environment of next-generation consumer services [10].

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First, static rule-based schemes cannot adapt to continuously evolving, cross-layer composite attacks and impose a substantial manual-maintenance burden. Second, traditional machine-learning models, while capable of detecting some anomalous traffic, have insufficient representational capacity for high-dimensional [11], heterogeneous data. Third, mainstream deep-learning detectors exhibit reduced adversarial robustness, limited adaptability, and compromised cross-layer correlations despite high accuracy. They lack core robustness design, effective mechanisms, and cross-layer intelligence fusion, failing to satisfy high precision, robustness, and low latency requirements for next-generation consumer services [12], [13].

Current AI intrusion detection for consumer services is vulnerable to adversarial attacks, noise, and concept drift. Multi-dimensional attack chains are covert and cross-layer correlated, impeding real-time accuracy and threat awareness. This study introduces robust detection via core reconstruction, cross-layer perception, and dynamic defense closed-loop. The main contributions of this paper are summarized as follows:

- Proposes a lightweight detection model integrating adversarial training and knowledge distillation, employing gradient masking and noise injection for robustness, and featuring a drift-aware module for adaptive stability.
- Proposes a heterogeneous GNN-based model for cross-layer attack-chain analysis, integrating edge-cloud feature extraction and adaptive task offloading to enable precise attribution with millisecond response.
- Swarm learning enables a distributed threat intelligence network for cross-domain knowledge aggregation, enhancing unknown threat detection. An uncertainty-estimation-driven hierarchical response dynamically triggers precise defensive actions, forming an autonomous detection-decision-suppression loop.

Here is the structural outline of the remaining content: Section II conducts a comprehensive review of existing literature; Section III introduces the comprehensive model and its underlying framework; Section IV elaborates on the experimental investigations on individual components; Section V presents the systematic analysis of performance metrics; whereas Section VI delivers the concluding remarks and future research directions.

II. RELATED WORK

Malware is a broad term referring to “malicious software.” Such software can gain access to or infect a system without the system owner’s knowledge. In this paper, we provide a brief overview of various known types of malware to date, including viruses, ransomware, botnets, adware, and others. In addition, we discuss the corresponding countermeasure—intrusion detection. Li et al. [14] proposed an accelerated attack correcting adversarial gradients with original images during backpropagation, showing superior transferability over state-of-the-art attacks against undefended and defended models. He et al. [15] introduced the momentum-enhanced PGD attack (M-PGD) and applied it in adversarial training, significantly enhancing model robustness. Huang et al. [16] developed an adaptive step-size method (ATAS) that effectively mitigates overfitting in FGSM adversarial training. Evaluations

on relevant datasets showed a marked improvement in model robustness.

Knowledge distillation trains a robust teacher from lightweight-generated samples, distilling robustness knowledge via probability distributions to a student model, ensuring inherited robustness and maintained efficiency. Wang et al. [17] addressed resource constraints of IoT devices by proposing a lightweight intrusion detection model, TBCLNN, which integrates bHHO-based feature dimensionality reduction, Tied Block Convolution (TBC), and an improved Self-Knowledge Distillation (SKD) loss. This approach achieves over 99% accuracy on multiple datasets while maintaining a compact model size and low computational cost. Yang et al. [18] introduced the LNet-SKD method to tackle network intrusion detection challenges in resource-limited environments. By leveraging the DeepMax module and self-knowledge distillation, LNet-SKD reduces parameter count and computational cost while outperforming existing techniques. Aljuhani et al. [19] proposed a lightweight intrusion detection framework that combines knowledge distillation, fuzzy logic, structured pruning, and quantization to balance security and efficiency on resource-constrained devices, achieving 98% accuracy with significantly reduced model resource overhead.

Next-generation consumer services, with multi-layer architectures and massive data interactions, face cross-layer attack vulnerabilities. A multi-dimensional attack-chain model employs spatiotemporal graph neural networks and endpoint-network dual-fusion for real-time threat propagation modeling. This approach adapts to dynamic data drift and personalized behaviors, enabling precise defense from point detection to global tracing, thereby ensuring system resilience and user security. Wang et al. [20] addressed limitations in early detection and transferability by proposing K-GetNID, which employs heterogeneous temporal graphs (HTGraph) and HTGNN models guided by knowledge to enable efficient intrusion detection. Hahn et al. [21] proposed a layered attack-chain framework for cyber-physical systems using kill-chain meta-model, enabling causal chain analysis and cross-layer attack tracing for multi-dimensional defense. Zhang et al. [22] proposed PdGAT-ID for ICS intrusion detection, integrating periodic features with spatiotemporal GAT, boosting F1 scores by 1.55–5.51% on SWaT.

Next-gen consumer services, reliant on IoT, cloud, and big data, face expanded attack surfaces. A decentralized threat intelligence network enables real-time multi-source analysis, dynamic updates, and cross-domain defense, enhancing APT and ransomware response while ensuring system resilience and data security. Homan et al. [23] introduces a decentralized CTI exchange architecture leveraging blockchain technology, with modularized communication pathways governed by the Traffic Light Protocol (TLP) framework. However, it lacks considerations for incentives, privacy, and confidentiality. Jiang et al. [24] addressed privacy concerns in network threat intelligence sharing by proposing the BFLS method, which combines blockchain and federated learning to achieve secure CTI sharing and high-accuracy threat detection. Juan et al. [25] proposed a framework integrating privacy-enhancing technologies with federated processing, addressing information

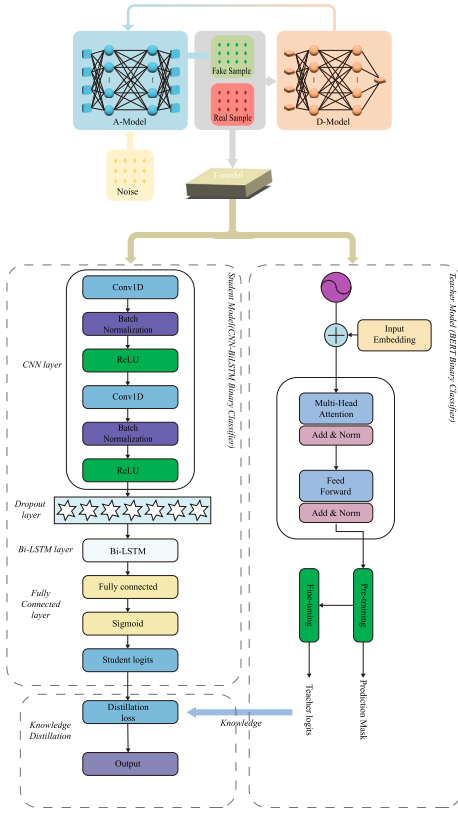


Fig. 1. Robust intrusion detection framework based on adversarial training and knowledge distillation.

asymmetry in CTI sharing, boosting dataset size and defense accuracy while balancing confidentiality and sharing benefits.

III. MODEL FRAMEWORK

A. Intrusion Detection Framework Integrating Adversarial Training and Knowledge Distillation

1) *Series Adversarial Training-Based Intrusion Detection Model Inspired by Generative Adversarial Networks (GANs)*: We propose an online knowledge distillation-based lightweight adversarial training model tailored for consumer service platforms. The model is designed to mitigate threats such as API abuse, which can cause service interruptions or performance degradation. It is applicable to training diverse intrusion detection systems (IDS), as illustrated in Figure 1.

The adversarial training adopts a framework inspired by Generative Adversarial Networks (GANs). To enhance the performance of the intrusion detection system (IDS), we employ an generator module (A-model) and a discriminator module (D-model) to generate adversarial samples, which are subsequently used to train the IDS, enabling it to resist attack samples effectively. Taking the distribution of normal samples in the training set as the target distribution, the A-model first receives random noise and maps it to samples that closely approximate the distribution of normal data. A dual-stage adversarial learning architecture is implemented, where the D-model computes the divergence between synthetic and legitimate data distributions. The A-model iteratively adjusts its parameters to minimize this divergence metric, forming an adversarial equilibrium.

A minimax game between A-model and D-model is formulated: generator amplifies discriminator activations for adversarial samples, while discriminator suppresses activations for synthetic data and enhances for authentic samples, establishing competitive equilibrium. The game between the A-model and the D-model can be expressed as:

$$\min_A \max_D E_{x \sim P_r} [D(x)] + E_{A(z) \sim P_g} [-D(A(z))] \quad (1)$$

Here, x denotes a real sample and z denotes a random noise vector. The outputs of the A-model and D-model are represented as $A(z)$ and $D(x)$, respectively. When the A-model is fixed, the loss function can be expressed as follows:

$$\text{loss}(x, z) = D(x) - D(A^*(z)) \quad (2)$$

Since $D(x)$ remains constant when the D-model is fixed, Equation (2) can be simplified to Equation (3). At this stage, the A-model should minimize Equation (3) to establish a minimax game between the a-model and the D-model.

$$\text{loss}(z) = -D^*(A(z)) \quad (3)$$

2) *A Knowledge Distillation Framework Based on BERT and CNN-BiLSTM*: The adversarially trained BERT-based binary classifier (teacher model) is compressed via lightweight knowledge distillation for efficient deployment on resource-constrained consumer service platforms. The framework integrates a BERT teacher, CNN-BiLSTM student, and distillation process, with BERT automatically extracting relevant features from network traffic to eliminate manual feature engineering for intrusion detection [26]. BERT, a pretrained deep neural network, substantially enhances intrusion detection efficiency and accuracy through large-scale pretraining. It offers strong cross-domain adaptability and fine-tuning capabilities for task-specific optimization. The core architecture employs a Transformer encoder with multi-head self-attention and feed-forward neural networks (FFNN) for effective feature representation.

a) *Multi-head self-attention mechanism*: Multi-head self-attention partitions input features into 8 parallel subspaces (optimal via ablation studies), projecting them into query (Q), key (K), and value (V) vectors. Each head computes attention weights via Q-K similarity and aggregates V for feature extraction. Outputs are concatenated and linearly transformed to fuse multi-perspective representations, enabling robust capture of local correlations and global dependencies for high-accuracy intrusion detection in network security systems. The computational formula is as follows:

$$\text{Attention}(Q, K, V) = \sigma \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (4)$$

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \text{head}_2, \dots, \text{head}_h) \cdot W^O \quad (5)$$

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V) \quad (6)$$

Here, σ denotes the softmax function, and Q , K , and V represent the query, key, and value matrices, respectively. The variable d_k indicates the dimension of the key vectors. Additionally, T denotes the transpose operation. The variable h represents the number of attention heads, while W_i^Q , W_i^K , W_i^V , and W^O are learnable parameter matrices.

b) *Feed-forward neural network (FFNN)*: The FFNN applies further transformations to the vector at each position. The model implementation includes a pair of affine operations combined with a rectified linear unit (ReLU) activation mechanism. This operation can be analytically described as:

$$FFNN(z) = \max(0, z \cdot W_1 + b_1) \cdot W_2 + b_2 \quad (7)$$

Here, z denotes the input to the FFNN layer. W_1 and b_1 represent the weight matrix and bias vector of the first fully connected (FC) layer, respectively, while W_2 and b_2 denote the weight matrix and bias vector of the second FC layer.

The output layer of the BERT model varies according to the specific task. During the pretraining phase, this layer is utilized to compute the loss for the masked language model (MLM).

$$\mathcal{L}_{MLM} = -\sum_{t \in M} \log P(x_t | X_{\setminus M}) \quad (8)$$

Here, M denotes the set of masked positions, and $X_{\setminus M}$ represents the input sequence containing masked tokens. The variable x_t corresponds to the true value of the masked word at index t .

Due to resource constraints on consumer service platforms, the student model reduces computational complexity and compresses size while preserving high intrusion detection performance. The CNN-BiLSTM architecture combines CNN for local feature extraction and BiLSTM for long-term dependency capture, significantly enhancing detection accuracy. [27]. As illustrated in Fig. 1, the model employs two CNN layers with Batch Normalization and ReLU for efficient feature extraction, enhancing stability and convergence. Dropout mitigates overfitting. A Bi-LSTM layer captures bidirectional dependencies, transforming spatial features to temporal. A fully connected layer with sigmoid outputs binary probabilities. This architecture integrates CNN, BN/ReLU, and Bi-LSTM for high-accuracy intrusion detection with minimal computational overhead on resource-constrained platforms.

B. Edge-Cloud Collaborative Multi-Dimensional Attack Chain Analysis

Based on the knowledge distillation framework, a multi-dimensional attack chain analysis model is developed for cross-layer threat tracing via edge-to-cloud architecture, as shown in Fig. 2.

Edge nodes transmit intrusion features to the cloud for global correlation analysis via a deep correlation engine. The cloud aggregates updated models, redistributes them with embedded attack patterns, and generates structured attack chain evidence for threat intelligence sharing. We propose STHGN for multi-dimensional attack chain analysis, which separately encodes semantic and structural features to effectively extract spatial information. Specifically, it constructs sentences based on each node's first-hop neighbors and employs Word2Vec to learn the semantic features of nodes. Structural features are represented via HGNN for spatial modeling. Temporal dependencies are modeled by TGN, and a multi-head self-attention mechanism fuses spatial and

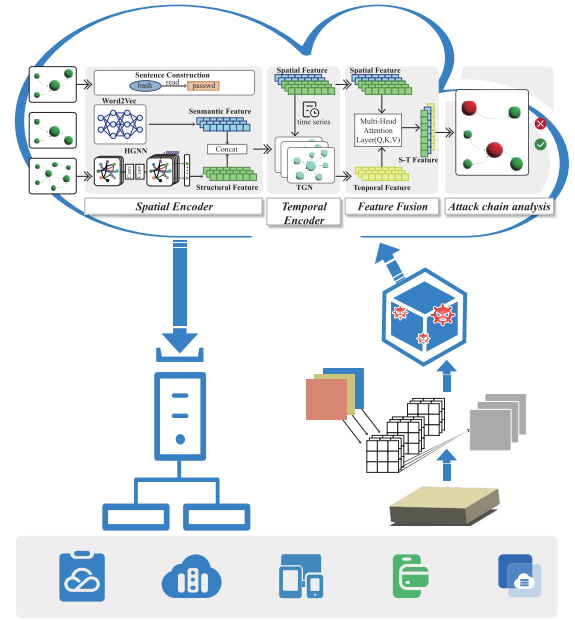


Fig. 2. Next-generation edge-cloud collaborative multi-dimensional attack chain analysis framework for consumer services.

temporal features to generate robust spatiotemporal embeddings, enabling comprehensive multi-dimensional attack chain analysis.

$$L = \sum_{w_t \in V} \sum_{-c \leq j \leq c, j \neq 0} \log P(w_{t+j} | w_t) \quad (9)$$

where V denotes the vocabulary, w_t is the target word, w_{t+j} is a context word, c represents the context window size, and $P(w_{t+j} | w_t)$ denotes the probability of predicting the context word w_t given the target word w_{t+j} . This probability is typically computed using the softmax function:

$$P(w_{t+j} | w_t) = \frac{\exp(v_{w_{t+j}}^T \cdot v_{w_t})}{\sum_{w \in V} \exp(v_w^T \cdot v_{w_t})} \quad (10)$$

Here, v_w represents the vector representation of w .

After semantic feature extraction, a Heterogeneous Graph Neural Network (HGNN) is employed to capture the structural characteristics of nodes. Unlike conventional Graph Convolutional Networks (GCNs), the HGNN [28] adopts a hierarchical processing strategy that combines local neighborhood information with global topological structures, allowing it to aggregate information according to the relative importance of different neighbors. In structural feature extraction, STHGN's HGNN employs a hierarchical aggregation mechanism with dual receptive fields based on one-hop neighbors and meta-paths to capture both local and global structural information. For a heterogeneous graph containing nodes such as users, devices, and IP addresses, HGNN first captures local structures (e.g., "user-device" direct connections) through the one-hop neighbor receptive field. Subsequently, it integrates multi-hop path information (e.g., "user-device-IP address") via the meta-path-based receptive field. This hierarchical processing strategy enables the model to dynamically adjust the receptive field size, preserving local details while capturing global topological patterns. During each aggregation layer, HGNN utilizes

graph convolution operations to perform weighted fusion of neighboring node information, ensuring critical connections' features are adequately preserved while suppressing the impact of noisy links. In the attack chain, nodes such as users, devices, and IP addresses contain diverse attributes, including user activity logs, device fingerprints, and geographical locations. Through hierarchical aggregation, the HGNN integrates these heterogeneous node features into a unified representation space.

Given a heterogeneous graph $\mathcal{G}$, the goal of the HGN is to process the message receptive fields $\mathcal{C}$. There are two types of message receptive fields: the first type is based on one-hop neighbors, which considers directly connected neighboring nodes during message passing. The second type is based on meta-paths, where a set of receptive fields $\mathcal{C}$ can be extracted according to predefined meta-path settings $\mathcal{T}_p$. For a given heterogeneous graph $\mathcal{G}$, let X^l denote the node representation at layer l , and $\mathcal{C}$ be the set of message receptive fields extracted from $\mathcal{G}$. Let t represent the target node. Then, the representation X_t^{l+1} at layer $(l+1)^{th}$ can be obtained by aggregating the messages from the neighbors X_s^l of t across all receptive fields in $\mathcal{C}$ at layer l , i.e.:

$$X_t^{l+1} \leftarrow \text{HAggr} \left(\text{Aggr}^{(\mathcal{C})} \left(\text{Encd}^{(\mathcal{C})} \left(X_s^l, X_t^l \right) \right) \right) \quad (11)$$

The above model can be divided into the following components:

$$M_{st}^{(\mathcal{C})} \leftarrow \text{Encd}^{(\mathcal{C})} \left(X_s^l, X_t^l \right) \quad (12)$$

$$X_t^{(\mathcal{C})} \leftarrow \text{Aggr}^{(\mathcal{C})} \left(\left\{ M_{st}^{(\mathcal{C})} \mid \forall (s \rightarrow t) \in \mathcal{C} \right\} \right) \quad (13)$$

$$X_t^{l+1} \leftarrow \text{HAggr} \left(\left\{ X_t^{(\mathcal{C})} \mid \forall \mathcal{C} \in \mathcal{C} \right\} \right) \quad (14)$$

In this formula, $(s \rightarrow t) \in \mathcal{C}$ denotes a neighboring node s of the target node t within a message receptive field $\mathcal{C} \in \mathcal{C}$. $\text{Encd}(\cdot)$ represents a message encoding function that extracts information from neighboring nodes X_s^l , encodes their representations, and generates the message $M_{st}^{(\mathcal{C})}$. $\text{Aggr}(\cdot)$ is a feature aggregation function that aggregates messages from neighbors within each receptive field $\mathcal{C} \in \mathcal{C}$, while $\text{HAggr}(\cdot)$ denotes a hybrid aggregation function that integrates the representations from all message receptive fields.

It is important to note that $\text{Aggr}(\cdot)$ and $\text{HAggr}(\cdot)$ share the same underlying operational function. The $\text{Aggr}(\cdot)$ function processes messages within a single receptive field $\mathcal{C}$, where each input corresponds to the representation of an individual neighboring node. In contrast, the $\text{HAggr}(\cdot)$ function aggregates messages across multiple receptive fields $\mathcal{C}$, where each input corresponds to the aggregated representation of a specific receptive field.

Given the inherent dynamism of cyberattacks, modeling temporal information in the source graph is critical for capturing evolving malicious activities. TGN [29] analyzes time-series features—such as transaction timestamps and login frequencies—to facilitate global temporal modeling. In the temporal embedding generation phase, each edge timestamp t_e is embedded into a vector t_e using positional encoding. This method maps the timestamp into a high-dimensional

space through sine and cosine functions. Formally, for a given timestamp t_e , the temporal embedding is computed as:

$$t_e = \left[\sin \left(\frac{t_e}{10000^{2i/d}} \right), \cos \left(\frac{t_e}{10000^{2i/d}} \right) \right]_{i=1}^{d/2} \quad (15)$$

Here, d denotes the dimensionality of the embedding space, and i denotes the index of the positional-encoding dimension.

In the time-aware information aggregation phase, the temporal feature $h_i(t)$ of each node v_i is updated by integrating information from its neighboring nodes. Specifically, the feature representation $h_i(t)$ of node v_i at time t is updated by aggregating the features of its neighbors along with the temporal embeddings of the edges connecting them. For a neighboring node v_j and an edge e_{ij} , the update rule is expressed as:

$$h_i(t) = \text{AGGREGATE} \left(\{ h_j(t) \| t_{e_{ij}} \mid v_j \in \mathcal{N}(v_i) \} \right) \quad (16)$$

In this equation, AGGREGATE represents the aggregation function, the symbol $\|$ denotes the feature concatenation operation, and $\mathcal{N}(v_i)$ denotes the set of neighboring nodes of node v_i .

Finally, during the node-embedding update phase, the embedding $h_i(t)$ of each node at the end of time step t is updated to a new time-aware embedding. The update function typically comprises a fully connected layer followed by an activation function, which integrates the previous time-step state with the current time-step features. The update equation is expressed as:

$$h_i(t+1) = \text{ReLU} (W \cdot h_i(t) + b) \quad (17)$$

Through these steps, the TGN generates time-aware node embeddings $h_i(t)$ that reflect the dynamic behavior and temporal evolution of nodes in the graph. A multi-head self-attention mechanism further enhances these time-aware embeddings to better capture long-term temporal dependencies.

To seamlessly integrate spatial features—derived from semantic vectors obtained via Word2Vec and structural features captured by the HGN—with temporal features extracted through the TGN, we employ a multi-head self-attention mechanism. This design choice aims to ensure that the fused spatiotemporal representations effectively capture temporal dependencies among nodes while retaining the rich information embedded in both spatial and temporal domains. By leveraging multi-head self-attention, the model can attend to diverse aspects of the spatiotemporal features from multiple perspectives, thereby generating more accurate and robust representations for multi-dimensional attack chain analysis and detection.

Formally, let the spatial vector X_s , which concatenates the Word2Vec-derived semantic features and the HGNN-extracted structural features, and the temporal vector X_t , obtained from the TGN, represent the inputs to the fusion stage. For each attention head h , we compute the query, key, and value matrices as follows:

$$\begin{aligned} Q_h &= W_{Q_h} [X_s \| X_t], \\ K_h &= W_{K_h} [X_s \| X_t], \\ V_h &= W_{V_h} [X_s \| X_t] \end{aligned} \quad (18)$$

where W_{Q_h} , W_{K_h} , and W_{V_h} are the learned projection matrices for head h , and $\parallel$ denotes concatenation.

Within the multi-head attention framework, the scaled dot-product mechanism is employed to derive attention weights for each individual head:

$$Attention_h(Q_h, K_h, V_h) = softmax\left(\frac{Q_h K_h^T}{\sqrt{d_k}}\right) V_h \quad (19)$$

where d_k denotes the dimensionality of the key vectors. These attention scores allow the model to assign varying weights to different spatiotemporal features, thereby enabling it to dynamically focus on the most relevant aspects.

$$Z = W_O \left[\parallel_{h=1}^H Attention_h(Q_h, K_h, V_h) \right] \quad (20)$$

where W_O denotes the learnable output projection matrix, and H denotes the number of attention heads.

The final output Z represents the fused spatiotemporal features, capturing diverse relationships across both temporal and spatial dimensions. The multi-head attention mechanism enhances the model's capacity to learn complex spatiotemporal patterns, thereby improving the performance of downstream attack detection tasks. By modeling long-term dependencies across time and space, this mechanism facilitates the detection of slow, continuously evolving cyberattacks, which is essential for identifying anomalies over extended time horizons.

C. Next-Generation Consumer Service Threat Intelligence Sharing Network (CTIS)

Next-generation consumer services, which integrate emerging technologies such as big data, cloud computing, and 6G, are confronted with increasingly severe cybersecurity challenges. To address these issues, a strategic defense framework emphasizing proactive defense, traceability, and adversarial resilience has been proposed [30]. Building on detection and analysis, a threat intelligence sharing network architecture is developed, primarily leveraging federated learning and blockchain technologies [31]. This framework enables secure CTI sharing via collaborative global model training, safeguarding confidential data. A trust-based evaluation framework employs auditing entities to assess and filter suboptimal local updates during aggregation, with credibility verified using zero-knowledge proof (ZKP) protocols for robust validation. Zero-knowledge proofs verify security feature authenticity without exposing sensitive data, balancing privacy and security in multi-device collaborative defense. The lightweight framework is feasible on resource-constrained consumer devices, which are high-value attack targets. This mechanism enhances detection and defense against advanced persistent threats while securing user data, providing a reliable privacy solution for next-generation consumer services. A visual representation of the complete system architecture is provided in Figure 4.

Edge node clusters function as federated learning clients, retaining private data and training models. Verifier nodes, selected by reputation scores, rank local contributions per iteration to identify the global model (GM), enabling secure CTI sharing. Verifier trustworthiness is validated via zero-knowledge proof (ZKP) protocols. Cluster aggregation nodes randomly aggregate top-quality models. Blockchain stores

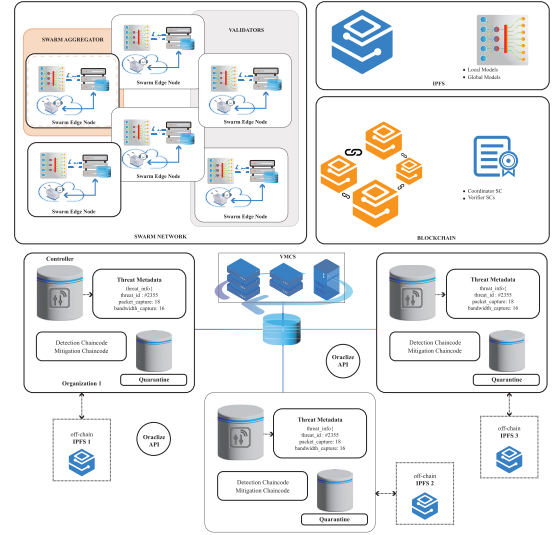


Fig. 3. Distributed threat intelligence sharing and defense framework based on federated learning and blockchain technologies.

ZKP outputs and transaction results. The Coordinator Smart Contract (CSC) manages verification addresses, IPFS model references, trust metrics, and iteration counts. The Verifier Smart Contract (VSC) processes ZKP proofs, ensuring robust network security through efficient CTI sharing. Smart contracts enable secure sharing and verification of threat intelligence through pre-defined automated rules. When a node submits threat intelligence, the smart contract first verifies its digital signature and data hash value to ensure that the intelligence source is trustworthy and has not been tampered with; Subsequently, the contract triggers a multi node verification mechanism based on PBFT consensus algorithm, requiring at least 70% of trusted nodes to cross verify the intelligence content. Only after the verification is passed, the intelligence will be written into the blockchain and broadcasted to the entire network. This mechanism protects node privacy through zero knowledge proof technology. A core component of the defense chain architecture is a blockchain-based consortium trust setup, implemented atop the Dolus defense mechanism as described in [32]. The defense chain evaluates peer-provided detection and mitigation services using Quality of Detection (QoD) and Quality of Mitigation (QoM) metrics, enabling requesters to select high-quality domains. The architecture integrates on-chain and off-chain components: IPFS serves as decentralized off-chain storage for threat intelligence, while blockchain anchors transactional records. Organizations share a common IPFS instance, with oracles facilitating periodic data interactions. On-chain chaincode retrieves IPFS-stored details to compute QoD/QoM, with storage capacity scaling dynamically to transaction volume and attack frequency. IPFS hashes are referenced in chaincode for efficient peer-to-peer data access. Service providers detect attacks, gather evidence, and deploy mitigations; completed transactions with comprehensive details are appended to blockchain, enabling secure, transparent CTI sharing among consortium members for robust network security.

Upon attacks targeting federated peers, the Freatic runtime-enabled monitoring subcomponent tracks cloud network traffic via SDN switches. The defensive architecture employs a

dual-phase ensemble learning framework (Dolus-inspired) to process traffic from malicious and legitimate perspectives. The first phase uses multivariate Gaussian methods for anomaly detection (e.g., connection exhaustion), identifying multi-source attack patterns to distinguish attackers from benign users, thereby enabling highly effective adversarial pattern categorization in real-time for enhanced network security. Outlier detection is performed using multiple effective multivariate detectors, each generating a set of hypotheses $\mathcal{H} = \{h_1, h_2, \dots, h_n\}$. The combined result corresponds to the hypothesis set described by the following equation. For improved accuracy, a Bayesian voting scheme is employed, which leverages the posterior probabilities of each hypothesis to assess their likelihoods.

$$\mathcal{F} = \sum_{h \in \mathcal{H}} h(x) P(h | D) \quad (21)$$

The final ensemble result $\mathcal{F}$ consists of all hypotheses in $\mathcal{H}$, with each hypothesis h weighted by its posterior probability $P(h | D)$. The posterior probability is proportional to the likelihood of the training data D given h multiplied by the prior probability of h .

$$P(h | D) \propto P(h) P(D | h) \quad (22)$$

Second-stage anomaly classification distinguishes significant events from false alarms via a simple classifier: consistent multi-threshold outcomes indicate adversarial patterns, divergent results are classified as benign. A suspicion score, combined with adversarial metadata (malicious entity, IP, traffic volume), is stored in IPFS. The detection chaincode computes and displays the QoD score on the UI, with provider peers initiating attack detection upon service requests. Upon receiving service requests, provider peers initiate attack detection. The attack detection time, combined with $t_{\min}$ (the assumed minimum time required to detect or mitigate attacks) and $t_{\max}$ (the assumed maximum time to detect or mitigate attacks), is used to calculate a normalized detection time, which in turn is utilized to estimate QoD. Let t'_d denote the normalized detection time based on the actual attack detection time t_d . It is defined as:

$$t'_d = \frac{t_d - t_{\min}}{t_{\max} - t_{\min}} \quad (23)$$

QoD estimate, prior to accounting for penalties and based on the deadline t_{deadline} and the actual response time t_r , is defined as:

$$A = \frac{1}{n} \sum_{i=1}^n \frac{a_i * \sum_{j=1}^k \frac{S_{\text{norm}}}{k}}{t'_d} \quad (24)$$

The final Quality of Detection (QoD) calculation treats the response time as the critical factor for score assignment. Let y represent the penalty variable defined by the Watchdog, which is applied when the response time exceeds the deadline. In this context, y is set to 0.8, corresponding to a 20 % reduction in the score. Expanding on Equation (25), the QoD is expressed as:

$$QoD = \begin{cases} A, & \text{if } t_r \leq t_{\text{deadline}} \\ A \cdot y, & \text{if } t_r > t_{\text{deadline}} \end{cases} \quad (25)$$

TABLE I
EXPERIMENTAL ENVIRONMENT

Category	Name	Details
Hardware	CPU	AMD Ryzen7 5800H 3.2GHz
	GPU	NVIDIA GeForce RTX 3070
	RAM	32G
Software	Programming	Python 3.8
	Platform	TensorFlow 2.6.0
	Tool	Wireshark, SplitCap2.1 and Scapy

Following attack detection, the requester submits a mitigation request based on trust metrics. The detection module synchronizes with mitigation for real-time threat characterization and response coordination. Candidate and trusted mitigators participate, with chaincode automating strategy execution. Attack traffic is routed to QVMs, resource availability verified, and response time reported. The chaincode analyzes system resources, service latency, and detection performance to compute Quality of Mitigation.

Let t denote the normalized time to mitigate an attack, scaled within the range $[0,1]$. Let er represent the attack recurrence rate, and S_r denote the success rate of the mitigator. The Quality of Mitigation (QoM) is defined by the following formula:

$$QoM = \frac{\sum_{i=1}^n S_r}{e^r \cdot t'_m} \quad (26)$$

IV. EXPERIMENT

A. Experiments on Lightweight Models Based on Adversarial Training and Knowledge Distillation

1) *Experimental Setup*: Performance evaluation of the proposed framework across three components—intrusion detection, multi-dimensional attack chain analysis, and threat intelligence sharing—used large-scale intrusion datasets. Hardware: AMD Ryzen 7 5800H, NVIDIA RTX 3070, 32GB RAM, Windows 10.

The software platform was implemented using Python 3.8, with TensorFlow 2.6.0 serving as the deep learning framework. Wireshark was employed to convert specific raw flow files into the desired format, while SplitCap (Version 2.1) was used for rapid segmentation of the raw flows. Flow processing within the system was performed using the Scapy Python package.

2) *Dataset*: Two public IoT datasets, NSL-KDD [32] and IoT-23 [33], were employed to evaluate the proposed intrusion detection model integrating adversarial training and knowledge distillation. Both datasets contain raw network traffic data, including diverse malicious traffic patterns from real IoT environments. The IoT-23 dataset, collected by the Stratosphere Laboratory of CTU University, comprises 23 traffic capture scenarios—20 from malware-infected IoT devices and 3 from benign IoT devices, with device names recorded.

3) *Evaluation Metrics*: To evaluate the performance of the proposed intrusion detection framework, we employed Accuracy, Precision, Recall, and F1-Score—metrics commonly used in intrusion detection system (IDS) evaluation. These metrics were computed as follows, where TP, FP, TN, and

TABLE II
SAMPLE NUMBER OF TWO DATASET CATEGORIES

Dataset	Category	Total Dataset
NSL-KDD	Normal	77054
	DoS	53385
	U2R	252
	R2L	3649
	Probe	14077
IoT-23	Normal	23791
	C&C	9498
	DDoS	15167
	C&C-Heartbeat	10872
	C&C-HeartbeatAttack	10661
	Okiru	11427
	PartOfAHorizontal	8691
	C&C-PartOfAH-PortScan	11245
	PortScan	9432

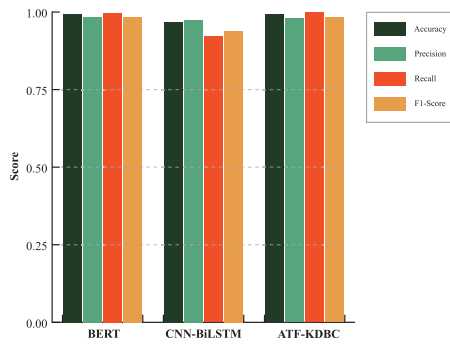


Fig. 4. Comparison of the three models on the NSL-KDD dataset.

FN represent true positives, false positives, true negatives, and false negatives, respectively.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (27)$$

$$Precision = \frac{TP}{TP + FP} \quad (28)$$

$$Recall = \frac{TP}{TP + FN} \quad (29)$$

$$F1 - Score = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (30)$$

4) *Experimental Results:* Experimental evaluation on two benchmark datasets confirms the ATF-KDBC framework's superior efficiency and accuracy over multiple baseline models, effectively demonstrating its enhanced performance for intrusion detection systems in network security applications.

Experiments on NSL-KDD and IoT-23 datasets compare BERT (teacher), CNN-BiLSTM (baseline), and ATF-KDBC against state-of-the-art methods. Figure 4 presents NSL-KDD results: BERT achieves 0.9936 accuracy and 0.9832 F1-score, while ATF-KDBC attains 0.9871 accuracy and 0.9823 F1-score, significantly outperforming baselines. This validates ATF-KDBC's enhanced intrusion detection efficacy for resource-constrained network security applications.

In contrast, the proposed ATF-KDBC model closely follows, exhibiting robust performance with an accuracy of 0.9925—slightly lower than that of the BERT model—while achieving a high F1-score of 0.9821. Notably, the original CNN-BiLSTM model, which lacks knowledge distillation optimization,

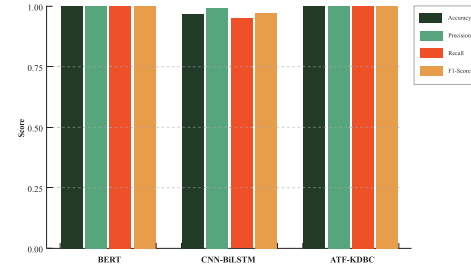


Fig. 5. Comparison of three models on the combined IoT-23 and NSL-KDD dataset.

TABLE III
THE COST OF THREE MODELS ON IoT-23 + NSL-KDD

Methods	Parameters	Model size(MB)
BERT	109483552	417.67
CNN-BiLSTM	615225	2.37
ATF-KDBC	265124	1.08

TABLE IV
DARPA E3 DATASET DESCRIPTION

Dataset	Edges	Nodes	Malicious	Size(GB)
Trace	4,080,457	3,220,596	68,082	18.6
Cadets	3,303,264	1,614,189	12,846	16.8
Theia	10,929,710	3,505,326	25,362	31.6
Total	18,313,431	8,340,111	106,290	67.0

demonstrates comparatively weaker detection performance among the three models, with an accuracy of 0.9687 and an F1-score of 0.9395.

Comparative analysis reveals that the knowledge distillation-enhanced KDBC model achieves performance nearly indistinguishable from the teacher BERT model while significantly outperforming the original CNN-BiLSTM model. Quantitatively, the ATF-KDBC model achieves an increase of 2.38 percentage points in classification accuracy and a rise of 0.0426 in F1-measure, thereby substantiating its superior performance and enhanced efficacy in adversarial attack detection.

To further validate the model's performance, this study combined the IoT-23 and NSL-KDD datasets and conducted experiments on the merged dataset. For the combined training set, the total training time was 3.36 h, and the average GPU utilization during training was 68%. The results, presented in Figure 5, systematically compare the key performance metrics of the three models. The teacher model (BERT) maintains the lead, achieving an accuracy of 0.9987 and an exceptionally high F1-score of 0.9987.

Following the BERT model is the proposed ATF-KDBC model, a CNN-BiLSTM architecture enhanced through adversarial training and knowledge distillation. The ATF-KDBC model exhibits strong performance on the combined dataset, achieving an accuracy of 0.9984 and an F1-score of 0.9985, with only a marginal difference from the teacher model. In contrast, the original CNN-BiLSTM model without adversarial training and knowledge distillation demonstrates comparatively weaker performance, with an accuracy of 0.9658 and an F1-score of 0.9696.

TABLE V
COMPARISON EXPERIMENTS

Model	Theia				Cadets				Trace			
	Precision	Accuracy	Recall	F1	Precision	Accuracy	Recall	F1	Precision	Accuracy	Recall	F1
HOLMES	0.0120	0.9961	0.9900	0.0230	0.0126	0.9557	1.0000	0.0245	0.0120	0.9825	1.0000	0.0220
Unicorn	0.6700	0.8000	0.6700	0.6700	0.3100	0.4400	1.0000	0.4700	0.2800	0.4300	1.0000	0.3400
AnoGraph	0.2319	0.9066	0.4619	0.3088	0.0606	0.8895	0.4698	0.2133	0.0294	0.7597	0.4698	0.0553
Threattrace	0.7156	0.9968	0.9999	0.8336	0.9035	0.9903	0.9997	0.9526	0.8701	0.9994	0.9963	0.9415
FLASH	0.9203	0.9944	0.9987	0.9519	0.9412	0.9998	0.9999	0.9605	0.9501	0.9989	0.9999	0.9703
TGN	0.9091	0.9967	1.0000	0.9524	0.8000	0.9940	1.0000	0.8889	0.8732	0.9911	1.0000	0.9284
GCN	0.8294	0.9844	0.9983	0.9060	0.8959	0.9925	0.9984	0.9444	0.9090	0.9937	0.9840	0.9381
STHGN	0.9487	0.9960	0.9987	0.9759	0.9863	0.9997	0.9987	0.9941	0.9950	0.9991	0.9883	0.9916

Comparative analysis of the experimental data indicates that the ATF-KDBC model not only approaches the performance of the teacher BERT model but also significantly outperforms the original CNN-BiLSTM model. Specifically, the ATF-KDBC model's accuracy increases by 3.26 percentage points, and its F1-score improves by 0.0289.

Due to limited computational resources on consumer platforms, lightweight models are evaluated for performance and size. BERT (417.67 MB) achieves superior accuracy but is bulky; CNN-BiLSTM (2.37 MB) and ATF-KDBC (1.08 MB) are compact, with ATF-KDBC offering optimal efficiency for resource-constrained intrusion detection deployments.

B. Experiment on Multi-Dimensional Attack-Chain Analysis

1) *Dataset*: The DARPA dataset [34] is a high-fidelity cybersecurity dataset crafted to simulate complex enterprise environments and advanced persistent threats (APTs). It emphasizes multi-stage attacks within realistic network architectures, capturing detailed host and network events encompassing both benign and malicious activities. Given our requirement for rich semantic information, we selected the more comprehensive Trace, Theia, and Cadets subsets for evaluation. Detailed dataset information is summarized in Table IV.

2) *Comparative Experiments*: In this section, we evaluate the performance of STHGN on three datasets and compare it with prior detection methods. Table V presents the experimental results, where STHGN consistently outperforms competitors across the Trace, Cats, and Theia datasets, achieving F1 scores exceeding 97%. By comparison, the static-feature-based detector HOLMES attains an F1 score of only 2.45%, the spatial-feature-based detector FLASH reaches up to 95%, and the temporal-feature-based detector TGN achieves approximately 95%. The early HOLMES detector, based on the APT lifecycle model [35], employs a set of TTP rules to identify patterns in the graph. However, these rules often match benign behavior, resulting in a high false positive rate and thus high recall but low precision (below 5%). Threattrace, relying solely on GraphSAGE for spatial features, neglects semantic and spatiotemporal information, resulting in poor detection ($F1 < 50\%$). FLASH enhances spatial feature extraction by combining Word2Vec semantic features with GCN structural features. Spatiotemporal detectors Unicorn and AnoGraph model temporal frequencies with low-dimensional features, yielding average $F1 < 50\%$ due to insufficient spatial context. The traditional TGN model outperforms them ($F1 > 90\%$) by effectively capturing node

spatial information. STHGN achieves the highest performance ($F1 > 98\%$) through integrated spatiotemporal feature fusion, generating highly discriminative representations for robust intrusion detection in network security applications.

V. CONCLUSION AND FUTURE WORK

This study proposes a resource-efficient intrusion detection framework based on adversarial training and knowledge distillation to address the limitations of existing algorithms in computational efficiency and robustness. The framework integrates an edge-cloud collaborative spatio-temporal heterogeneous graph network (STHGN) to analyze multidimensional attack chains, ensuring data privacy preservation and mitigating poisoning risks. Experimental results demonstrate that the proposed architecture achieves significant computational resource savings while enhancing detection and defense robustness. This work provides a robust security solution for next-generation consumer service platforms, establishing its practical applicability through quantitative evaluations. The methodology advances state-of-the-art intrusion detection by combining adversarial resilience mechanisms with distributed threat intelligence sharing, offering critical implications for secure consumer electronics ecosystems.

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